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Exp No: 12	**End-End Communication at Transport Layer - Echo Client Server and Chat**

AIM:

To implement (a) an echo client-server using TCP and UDP sockets and (b) a chat program using socket programming, for end-to-end communication at the transport layer.

ALGORITHM / PROCEDURE:

(a1) TCP echo server and client

1. **Server**: create a TCP socket with `socket(AF_INET, SOCK_STREAM)`, `bind()` it to an IP address and port (5000), call `listen()` and wait in `accept()` for the client.
2. **Client**: create a TCP socket and call `connect()` with the IP address and port of the server.
3. The client sends a message using `send()`. The server reads it with `recv()`, displays it and sends the same data back using `sendall()`.
4. The client receives the echoed message and displays it. This is repeated for each message.
5. Close the sockets using `close()`.

(a2) UDP echo server and client

1. **Server**: create a UDP socket with `SOCK_DGRAM` and `bind()` it to port 6000. No `listen()` / `accept()` is needed because UDP is connectionless.
2. **Client**: create a UDP socket and send the message to the server address with `sendto()`.
3. The server receives the datagram and the client address with `recvfrom()` and sends it back using `sendto()`.
4. The client receives the echo with `recvfrom()` and displays it. Close the sockets.

(b) Chat program

1. The chat server creates a TCP socket on port 7000 and waits for a client. The client connects to it.
2. The client types a message and sends it. The server displays it, then the server user types the reply and sends it back.

Point	TCP	UDP
Socket type	SOCK_STREAM	SOCK_DGRAM
Connection	Connection oriented (3-way handshake)	Connectionless
Reliability	Reliable, ordered delivery	No guarantee
Server calls	bind, listen, accept, recv, send	bind, recvfrom, sendto
Used in	HTTP, FTP, SMTP	DNS, DHCP, streaming

3. Messages are exchanged alternately until either side sends the word `exit`. Then both sockets are closed

echo_tcp_client.py

```
import socket c = socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
c.connect(("127.0.0.1", 5000)) for msg in ["Hello Server",
"Computer Networks Lab", "bye"]:
```

DIAGRAM:

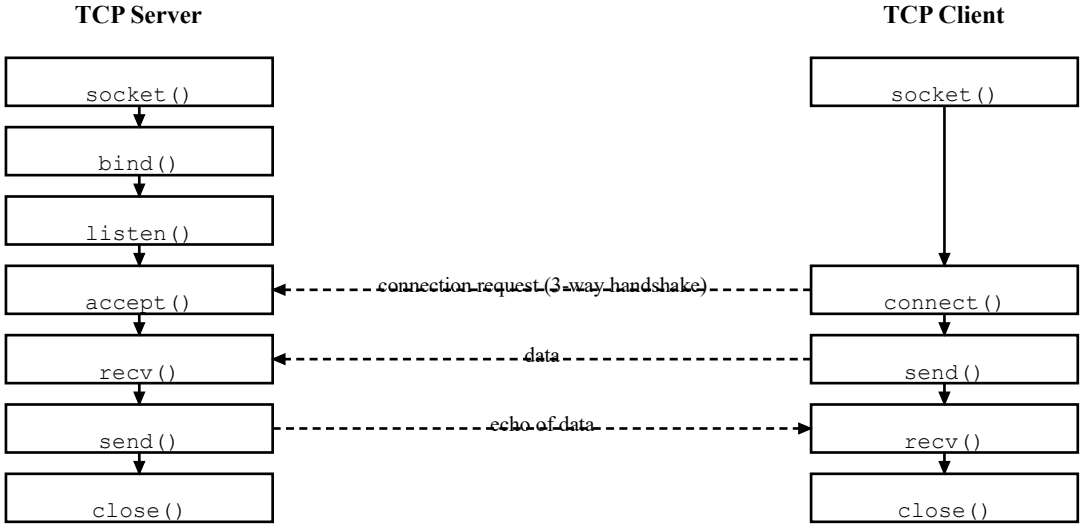


Fig 1: Socket calls in TCP client-server communication

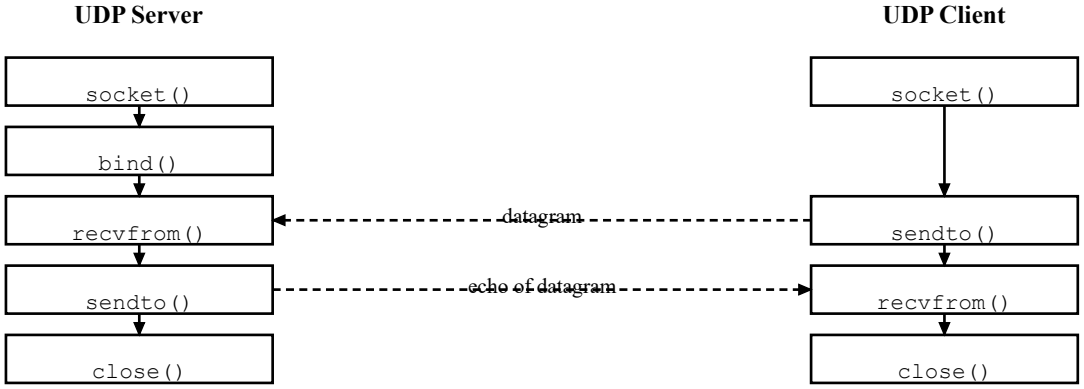


Fig 2: Socket calls in UDP client-server communication

PROGRAM:

echo_tcp_server.py

```
import socket s = socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
s.bind(("127.0.0.1", 5000))
s.listen(1) print("TCP Echo Server listening
on port 5000 ...") conn, addr = s.accept()
print("Connected by", addr) while True:
data =
conn.recv(1024)     if not
data:         break
    print("Received :", data.decode())
conn.sendall(data)               # echo
back conn.close() s.close()
print("Connection closed.")
```

echo_tcp_client.py (contd.)

```
    c.sendall(msg.encode())     print("Sent
:", msg)     print("Echoed   :",
c.recv(1024).decode())
c.close()
```

echo_udp_server.py

```
import socket s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
s.bind(("127.0.0.1", 6000)) print("UDP Echo
Server waiting on port 6000 ...") for _ in
range(2):     data, addr = s.recvfrom(1024)
print("Received from", addr, ":", data.decode())
s.sendto(data, addr)
s.close()
```

echo_udp_client.py

chat_client.py

```
import socket c = socket.socket()
c.connect(("127.0.0.1", 7000)) print("Connected to
chat server. Type 'exit' to quit.") while True:
msg = input("Client : ")
    c.send(msg.encode())
if msg.lower() == "exit":
break
    reply = c.recv(1024).decode()
if reply.lower() == "exit":
break     print("Server :",
reply) c.close() print("Chat
ended.")
```

```
import socket c = socket.socket(socket.AF_INET,
socket.SOCK_DGRAM) for msg in ["Hello UDP", "Datagram
test"]:
    c.sendto(msg.encode(), ("127.0.0.1",
6000))     data, _ = c.recvfrom(1024)
print("Sent     :", msg)     print("Echoed
:", data.decode()) c.close()
```

chat_server.py

```
import socket s = socket.socket()
s.bind(("127.0.0.1", 7000))
s.listen(1) print("Chat server started. Waiting
for client ...") conn, addr = s.accept()
print("Client connected:", addr) while True:
msg = conn.recv(1024).decode()     if not msg
or msg.lower() == "exit":         break
print("Client :", msg)     reply =
input("Server : ")
conn.send(reply.encode())     if reply.lower()
== "exit":         break conn.close() s.close()
print("Chat ended.")
```

OUTPUT:

(a1) TCP echo

userUSER @userUSER -VirtualBox: ~ (Terminal 1 - Server)

```
userUSER @userUSER -VirtualBox:~$ python3 echo_tcp_server.py TCP
Echo Server listening on port 5000 ...
Connected by ('127.0.0.1', 46248)
Received : Hello Server
Received : Computer Networks Lab
Received : bye
Connection closed.
```

userUSER @userUSER -VirtualBox: ~ (Terminal 2 - Client)

```
userUSER@userUSER -VirtualBox:~$ python3 echo_tcp_client.py
Sent     : Hello Server
Echoed   : Hello Server
Sent     : Computer Networks Lab
Echoed   : Computer Networks Lab
Sent     : bye
Echoed   : bye
```

(a2) UDP echo

userUSER @userUSER -VirtualBox: ~ (Terminal 2 - Chat Client)

```
userUSER @userUSER -VirtualBox:~$ python3 chat_client.py
Connected to chat server. Type 'exit' to quit.
Client : Hi Server, this is userUSER Server :
Hello userUSER , how can I help?
Client : Is the CN lab record ready?
Server : Yes, the lab record is due on Friday.
Client : Thanks, bye Chat
ended.
```

userUSER @userUSER -VirtualBox: ~ (Terminal 1 - Server)

```
userUSER @userUSER -VirtualBox:~$ python3 echo_udp_server.py UDP
Echo Server waiting on port 6000 ...
Received from ('127.0.0.1', 53205) : Hello UDP
Received from ('127.0.0.1', 53205) : Datagram test
```

userUSER @userUSER -VirtualBox: ~ (Terminal 2 - Client)

```
userUSER@userUSER -VirtualBox:~$ python3 echo_udp_client.py
Sent     : Hello UDP
Echoed   : Hello UDP
Sent     : Datagram test
Echoed   : Datagram test
```

(b) Chat program

userUSER @userUSER -VirtualBox: ~ (Terminal 1 - Chat Server)

```
userUSER @userUSER -VirtualBox:~$ python3 chat_server.py
Chat server started. Waiting for client ... Client
connected: ('127.0.0.1', 43260) Client : Hi Server,
this is userUSER Server : Hello userUSER , how can I help?
Client : Is the CN lab record ready?
Server : Yes, the lab record is due on Friday.
Client : Thanks, bye
Server : exit Chat
ended.
```

RESULT:

Thus the echo client-server programs using TCP and UDP sockets and the chat program using socket programming were implemented and executed successfully.